

optionwright

The LLM proposes.
The code decides.

THE IDEA

An LLM is good at reading context and **bad at discipline**. So we split the two.

Each cycle, deterministic code reads the option chain, filters for liquidity, and pre-builds the candidate spreads with every number already computed. The model returns one thing: a direction, or an abstention.

chain → filter + build spreads → LLM: bullish · bearish · abstain →
code: strikes · size · gates · execute

The model gets to have an opinion. **It never decides how much money is on the line.**

STRATEGY

Defined-risk vertical spreads only

- ❑ **Bull put** when bullish, **bear call** when bearish — credit spreads with a capped max loss (width – credit).
- ❑ Max loss is **fixed the moment the order fills**. No path to an unbounded loss.
- ❑ Liquid weeklies on **SPY, QQQ, IWM**; strikes by delta; 3–5 days to expiration.
- ❑ Exits by rule: **trailing take-profit**, stop-loss, forced close before expiration.

Losses bounded by construction – the risk profile is chosen, not hoped for.

RISK ENGINE

Eight gates that can veto or shrink — **never enlarge**

BEFORE EVERY ORDER

- ❑ Consecutive-loss breaker
- ❑ Open-positions cap + **per-underlying cap**
- ❑ Opening + macro blackout
- ❑ Max loss per position · daily budget

THE KEY PROPERTY

- ❑ Size **emerges from the gates**, not from the model
- ❑ Fail-closed: a bad LLM response **abstains**
- ❑ Every decision + veto **logged and auditable**

Deterministic guardrails matter more than any model swap.

UNDER THE HOOD

Portable by design, observable end to end

- ❑ **Alpaca Trading API + CLI** — chain, quotes, greeks, and multi-leg (mleg) execution.
- ❑ **Qwen 72B via Featherless** as the brain, with a **local Ollama fallback** — any OpenAI-compatible endpoint, no code change.
- ❑ **Postgres** for every decision + position; **Prometheus + Grafana** for live metrics.
- ❑ **66 tests** over the deterministic core. Judges run the whole stack with **docker compose up**.

Same code drives a 9B on my GPU or a 72B in the cloud — resilience, not a rewrite.

SEE IT LIVE

The agent decides in the open

A public dashboard shows the equity curve, open and closed spreads, and the **decision log** — the reason and the gate verdict behind every cycle. No black box.

LIVE DASHBOARD

optionwright.richardx.dev

REPO (MIT)

github.com/ricardogs26/hackathon-alpaca

Paper account · \$100,000 start · results updated through the judged week

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